

Data Quality Analysis

Identifying Factors Behind Low-Quality Data Records

Google Data Analytics Professional Certificate — Capstone Case Study

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Tools: Excel/Google Sheets • SQL • Python • Tableau • GitHub • Azure Machine Learning

Executive Summary

This case study analyzes a 50,000-record operational dataset created for an Azure Machine Learning Data Quality Agent. The business problem is the cost and risk of allowing low-quality records to reach analysts and downstream business users. The analysis examines which record characteristics are associated with a manual-review requirement and translates those findings into practical recommendations.

The dataset contains 5,112 records labeled as requiring review, representing 10.22% of all records. The analysis found clear relationships between review status and several quality indicators. Records with format errors had a 26.24% review rate compared with 8.44% for records without format errors. Records marked as duplicates had a 24.78% review rate compared with 8.93% for non-duplicates. Higher missing-value counts were also associated with higher review rates, while lower review scores were associated with a greater likelihood of review.

The analysis supports a targeted quality-control strategy: prioritize records with format errors, duplicate flags, high missing-value counts, and low review scores for earlier inspection. These findings also provide the analytical foundation for the machine-learning model developed in the related Azure ML project.

1. ASK — Define the Business Problem

Business Scenario

A company processes large volumes of task records before those records are delivered to analysts and business users. Some records contain quality problems such as missing values, duplicate records, formatting errors, unusual response characteristics, low review scores, or elevated processing times. When these records are not identified early, analysts may spend unnecessary time cleaning data, investigating questionable records, or making decisions from unreliable information.

The company wants to understand the patterns behind records that require review and determine which characteristics should receive the most attention during quality control.

Stakeholders

- Data analysts and business analysts who depend on reliable records.
- Data-quality or operations teams responsible for reviewing questionable records.
- Managers responsible for review workload, productivity, and data reliability.
- Technical teams responsible for implementing automated quality checks.

Business Question

What characteristics are most strongly associated with low-quality records, and how can the organization use those findings to improve its data-review process?

Analytical Questions

1. What percentage of records require manual review?
2. Which task types have the highest review rates?
3. How does the presence of duplicate records affect the likelihood of review?

4. How do formatting errors affect the likelihood of review?
5. How does the number of missing values relate to review status?
6. How does review score differ between records that do and do not require review?
7. Are response length and processing latency meaningfully different between the two groups?
8. Which factors should receive the highest priority in a data-quality workflow?

Success Criteria

The analysis is successful if it produces clear, reproducible findings that a business stakeholder can use to prioritize data-quality controls. The final deliverable should communicate the problem, methods, findings, visual evidence, and recommended actions without requiring the stakeholder to understand the underlying machine-learning implementation.

2. PREPARE — Understand the Data

Dataset Overview

Item	Description
Records	50,000
Features	7 input attributes
Target	needs_review (binary)
Positive records	5,112
Positive-class rate	10.22%
Data type	Structured tabular data

Variables

Variable	Description	Role
task_type	Type of task being processed	Categorical feature
missing_value_count	Number of missing values in the record	Numeric feature
duplicate_flag	Indicates whether the record is a duplicate	Binary feature
format_error_flag	Indicates whether a formatting problem exists	Binary feature
response_length	Length of the response	Numeric feature
review_score	Quality/review score associated with the record	Numeric feature
latency_seconds	Processing time in seconds	Numeric feature
needs_review	Indicates whether the record requires review	Target

Data Source and Scope

The dataset was developed for the related Data Quality Agent project and contains 50,000 synthetic operational records. It was designed to represent common data-quality conditions while providing a known binary outcome for analytical and machine-learning evaluation. Because the data are synthetic, the findings should be presented as a demonstration of analytical workflow rather than as claims about a real company's production data.

Data Quality Considerations

- The target is imbalanced: only 10.22% of records require review.
- Accuracy alone can therefore be misleading when evaluating predictive performance.
- Binary flags must be interpreted as indicators of observed record conditions, not proof of causation.
- Synthetic data can demonstrate methodology but should not be generalized to an actual production population without validation.

3. PROCESS — Clean and Prepare the Data

Cleaning Approach

The data-processing stage focused on ensuring consistent field types, checking for missing or invalid values, confirming the binary target representation, and verifying that the dataset could be analyzed consistently in Python and SQL. Categorical task types were retained for grouped analysis and encoded when required by machine-learning workflows.

- Verified the dataset contained 50,000 records.
- Verified the expected feature columns and binary target were present.
- Checked numeric fields for valid ranges and consistency.
- Reviewed duplicate and format-error indicators as explicit binary fields.
- Prepared categorical task_type values for grouped analysis and modeling.
- Preserved the needs_review target for supervised analysis.
- Used a stratified modeling workflow because the positive class represented only about one-tenth of the data.

Analytical vs. Machine-Learning Processing

For the Google Data Analytics capstone, the emphasis is on understanding the data and producing business insights. The related Azure ML project extended this process into predictive modeling. The analytical case study therefore uses the same prepared dataset while keeping the business question focused on descriptive and diagnostic analysis.

4. ANALYZE — Explore Patterns and Relationships

Overall Review Rate

Out of 50,000 records, 5,112 were labeled as requiring review and 44,888 did not require review. This means 10.22% of the records required review. In a workflow where every record is reviewed manually, approximately nine out of ten records would not need intervention based on the target definition. This creates an opportunity to prioritize review resources rather than treating every record equally.

Task Type Analysis

Task Type	Records	Needs Review	Review Rate
Safety Review	12,522	1,643	13.12%
Data Labeling	12,430	1,169	9.40%
Image Evaluation	12,432	1,143	9.19%
Text Evaluation	12,616	1,157	9.17%

Safety Review had the highest review rate at 13.12%, compared with approximately 9.17%–9.40% for the other task types. This difference suggests task type may be useful for prioritization, although task type alone does not explain most review decisions.

Duplicate Records

Duplicate Flag	Records	Review Rate
No	45,908	8.93%
Yes	4,092	24.78%

Records marked as duplicates had a 24.78% review rate, compared with 8.93% for records without the duplicate flag. The review rate was therefore substantially higher among duplicate records. This supports treating duplicate detection as an important early quality-control step.

Formatting Errors

Format Error Flag	Records	Review Rate
No	44,997	8.44%
Yes	5,003	26.24%

Formatting errors showed one of the clearest relationships in the dataset. Records with a format-error flag had a 26.24% review rate, compared with 8.44% when no format error was present. This makes format validation a strong candidate for automated pre-screening.

Missing Values

Missing Values	Records	Review Rate
0	5,491	4.99%
1	12,236	6.88%
2	13,438	8.92%
3	9,821	12.04%
4	5,366	14.96%
5+	3,648	22.29%

The review rate increased as the number of missing values increased. Records with no missing values had a 4.99% review rate, while records with five or more missing values had a 22.29% review rate. This pattern indicates that missingness is an important quality signal and that records with extensive missing data should receive higher review priority.

Review Score

Review Score Range	Review Rate
Below 60	28.12%
60–69	17.30%
70–79	12.28%
80–89	8.08%
90–100	4.71%

Lower review scores were associated with substantially higher review rates. Records with scores below 60 had a 28.12% review rate, compared with 4.71% for scores from 90 to 100. This inverse relationship provides a practical quality signal: lower-scoring records deserve greater scrutiny.

Response Length and Processing Latency

Metric	No Review	Needs Review
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Average response length	180.19	180.62
Average latency (seconds)	94.88	97.91

Response length was nearly identical between the two groups, suggesting that it is not a strong standalone descriptive separator in this dataset. Average latency was somewhat higher among records requiring review, but the difference was comparatively modest. These variables may still contribute useful information when combined with stronger quality indicators.

Combined Quality Signals

The strongest practical pattern appears when multiple quality indicators occur together. Among records with neither a duplicate flag nor a format-error flag, the review rate was 7.26%. When both a duplicate flag and a format-error flag were present, the review rate increased to 53.81%. This demonstrates why a multi-factor quality-control process can be more useful than relying on a single rule.

Key Analytical Findings

- 10.22% of all records require review, creating a meaningful opportunity to prioritize quality-control resources.
- Safety Review has the highest task-type review rate at 13.12%.
- Duplicate records have a 24.78% review rate versus 8.93% for non-duplicates.
- Records with format errors have a 26.24% review rate versus 8.44% without format errors.
- Review rates rise steadily as missing-value counts increase, reaching 22.29% for records with five or more missing values.
- Lower review scores are strongly associated with higher review rates: 28.12% below 60 versus 4.71% at 90–100.
- Response length shows little descriptive difference between the groups, while latency shows a smaller difference.
- Multiple quality flags together can identify a much higher-risk subset than any single indicator alone.

5. MODELING CONNECTION — From Analysis to Prediction

The descriptive analysis provided the rationale for the related machine-learning Data Quality Agent. The modeling objective was to predict `needs_review` so that potentially problematic records could be prioritized before manual review.

Because only 10.22% of records required review, class imbalance was a central modeling challenge. A baseline logistic-regression model achieved high accuracy but low recall, meaning it missed many records that actually required review. A class-weighted model substantially improved recall. Threshold tuning and hyperparameter optimization were then evaluated to balance precision and recall more appropriately for a quality-control use case.

Model	Accuracy	Precision	Recall	F1
Baseline Logistic Regression	90.23%	63.80%	10.18%	17.55%
Class-Weighted Logistic Regression	71.99%	21.63%	66.34%	32.62%

Threshold-Tuned Logistic Regression	86.15%	33.95%	37.48%	35.63%
Hyperparameter-Tuned Logistic Regression	86.15%	33.95%	37.48%	35.63%

The final evaluation used an untouched test set with a 0.70 decision threshold. The resulting accuracy was 86.16%, precision was 33.95%, recall was 37.48%, and F1 score was 35.63%. The confusion matrix was 8,233 true negatives, 745 false positives, 639 false negatives, and 383 true positives. Because the business objective emphasizes finding records that need attention, recall and F1 are more informative than accuracy alone.

6. SHARE — Communicate the Results

Recommended Tableau Dashboard

The final Tableau dashboard should be designed for a manager or data-quality lead. It should tell a simple story from overall workload to the strongest quality drivers.

- KPI 1: Total Records — 50,000.
- KPI 2: Records Requiring Review — 5,112.
- KPI 3: Review Rate — 10.22%.
- KPI 4: Highest-Risk Task Type — Safety Review at 13.12%.
- Bar chart: Review rate by task type.
- Bar chart: Review rate with versus without duplicate flags.
- Bar chart: Review rate with versus without format errors.
- Line or column chart: Review rate by missing-value count.
- Column chart: Review rate by review-score range.
- Comparison chart: Average latency and response length by review status.
- Optional combined-risk view showing review rate for duplicate/format-error combinations.

Suggested Dashboard Story

9. Start with the overall 10.22% review rate to establish the size of the problem.
10. Show which task type has the highest review rate.
11. Highlight format errors and duplicate records as strong quality signals.
12. Show the upward trend in review rate as missing values increase.
13. Show the inverse relationship between review score and review rate.
14. Finish with recommended actions and explain how the predictive model can prioritize records for review.

Presentation Summary

The business does not need to treat all 50,000 records equally. Only 10.22% require review, and several measurable characteristics are associated with a substantially higher review rate. Format errors, duplicate records, high missing-value counts, and low review scores are the clearest signals in the analysis. By using these indicators in a prioritized quality-control workflow, the organization can focus human attention where it is most likely to be useful.

7. ACT — Recommendations

Recommendation 1: Prioritize Format Validation

Implement automated format checks before records enter downstream analyst workflows. Records with formatting errors should be routed to a higher-priority review queue because their observed review rate is 26.24%.

Recommendation 2: Strengthen Duplicate Detection

Run duplicate checks early in the pipeline and prioritize flagged records. Duplicate records show a 24.78% review rate, substantially above the 8.93% rate among records without the flag.

Recommendation 3: Create Missing-Value Thresholds

Use escalating review priority as missing-value counts increase. Records with five or more missing values have a 22.29% review rate, compared with 4.99% when no values are missing.

Recommendation 4: Use Review Score as a Triage Signal

Give lower-scoring records higher review priority. The review rate is 28.12% for scores below 60 versus 4.71% for scores from 90 to 100.

Recommendation 5: Combine Signals Instead of Using One Rule

Use multiple quality indicators together. The combined duplicate-and-format-error condition produced a 53.81% review rate in the dataset, demonstrating that risk can increase substantially when multiple issues occur in the same record.

Recommendation 6: Use Machine Learning for Prioritization

Use the Data Quality Agent as a prioritization layer rather than as a replacement for human judgment. The model can assign a review probability and route higher-risk records to analysts while allowing lower-risk records to proceed through a lighter-touch workflow.

8. LIMITATIONS

- The dataset is synthetic and was created for demonstration and machine-learning development; it does not represent a verified production population.
- Associations do not prove that a specific factor causes a record to require review.
- The predictive model has a moderate F1 score and should therefore be treated as a prioritization tool rather than an autonomous final decision-maker.
- False negatives remain important because missed quality issues can reach downstream users.
- Production deployment would require monitoring, feedback from reviewers, recalibration, and validation on real historical data.
- Additional analysis could test whether the observed relationships remain stable over time and across different task types or operational groups.

9. CONCLUSION

This case study demonstrates how a data analyst can move from a business problem to actionable insight using a structured analytics process. The analysis of 50,000 records identified meaningful patterns in duplicate flags, formatting errors, missing values, review scores, and task types. These findings provide a practical framework for prioritizing quality-control work and directly informed the related machine-learning system.

The most important business takeaway is that data-quality review does not need to be treated as an all-or-nothing process. By identifying records with stronger quality-risk signals and prioritizing them for human attention, an organization can create a more efficient review workflow while improving confidence in the data delivered to analysts and decision-makers.